

PEDIATRIC CARDIOVASCULAR DISEASE

Outflow obstruction (well child)

Aortic stenosis

- Narrowing of valve from LV → aorta. **Valvular, subvalvular, or supravalvular** (~5%).
- **Clinical (depends on severity):** mild–moderate = asymptomatic; **severe = easy fatigability, exertional chest pain, syncope.**
- **Critical infant AS:** survives only if **PDA permits flow** to aorta + coronaries (duct-dependent).

Pulmonary stenosis

- Outflow obstruction RV → pulmonary artery. Mild = asymptomatic; severe → RV overload/failure. [Standard: ejection systolic murmur left upper sternal edge, balloon valvuloplasty if severe.]

Coarctation of aorta

- Narrowing of aorta (juxtaductal). [Standard: **radio-femoral delay, weak femoral pulses, upper limb hypertension, rib notching**; duct-dependent in neonate → **prostaglandin to keep PDA open**; surgery/balloon.]

Patent ductus arteriosus (PDA)

- Persistent ductus → left-to-right shunt. [Standard: **continuous "machinery" murmur**, bounding pulses; treat indomethacin/ibuprofen (close) or surgical/device closure.]

Heart Failure

Definition & physiology

- State where heart can't deliver adequate **cardiac output** for metabolic needs.
- **CO = HR × SV**. SV determinants: **preload (volume), afterload (pressure), contractility.**

Causes

- **Rhythm:** complete heart block, SVT, VT, sinus node dysfunction.
- **Volume overload:** structural (**VSD, PDA, aortic/mitral regurgitation**), anaemia, sepsis.
- **Pressure overload:** aortic/pulmonary stenosis, coarctation, hypertension.
- **Systolic dysfunction:** myocarditis, dilated cardiomyopathy, malnutrition, ischaemia.
- **Diastolic dysfunction:** hypertrophic/restrictive cardiomyopathy, pericarditis, **tamponade.**

Clinical

- **Infants:** feeding difficulty + sweating, poor weight gain, irritability, weak cry, respiratory distress.
- **Children:** fatigue, effort intolerance, anorexia/abdominal pain, dyspnoea, cough, orthopnoea.
- **Signs:** respiratory distress, ↑JVP, **hepatomegaly**, oedema, basal crepitations, cardiomegaly, **gallop rhythm**, holosystolic murmur (mitral/tricuspid insufficiency).

Investigations

- **CXR** (cardiomegaly, pulmonary vascularity), **ECG** (hypertrophy, ischaemia, rhythm), **echo** (ventricular function), blood gas (acidosis), electrolytes (hypoNa, hypoglycaemia).

Treatment

- Treat underlying cause. General: rest, **semi-upright (children) / infant chair**, modify activity.
- **Diet:** ↑calories (up to 24 cal/oz), **no added salt.**

- **Medications: diuretics, inotropes, afterload reducers (vasodilators).**

Kawasaki Disease

- Group of symptoms/signs; **children <5 yrs**. Cause unknown (?infection/toxin/immune). Low family recurrence.
- **Diagnosis: fever ≥ 5 days + 4 of 5:**
 1. **Non-purulent red eyes** (bilateral conjunctivitis)
 2. **Lip/tongue/mouth changes** (red cracked lips, **strawberry tongue**)
 3. **Hand/feet changes** (red/swollen palms+soles → **peeling fingers/toes ~1 wk after fever**)
 4. **Body rash** (red, bumpy; groin/diaper area, peels)
 5. **Cervical lymphadenopathy** (usually unilateral, $>1.5\text{cm}$)
- **Treatment: aspirin + IVIG** ($\pm$ steroids). No single cure.
- **Complications: coronary artery changes + aneurysms** (the danger).
- **Prognosis: 95% recover completely**; if no coronary damage on echo → complete recovery likely.

Cardiac Arrhythmias

Supraventricular tachycardia (SVT)

- **Most common childhood arrhythmia.** HR **250–300 bpm**. → poor CO, pulmonary oedema. Presents as **HF in neonate/infant**; can cause **hydrops fetalis + intrauterine death**.
- **Re-entry** via accessory pathway. Usually no structural disease (but echo to check).
- **ECG:** narrow-complex tachycardia 250–300; P wave after QRS (retrograde). **WPW:** short PR + **delta wave** + wide QRS (early antegrade conduction).
- **Management (severely ill → restore sinus rhythm):**
 1. Circulatory/respiratory support, correct acidosis.
 2. **Vagal manoeuvres** (carotid massage, **ice pack to face**) — ~80% success.
 3. **IV adenosine = treatment of choice** (induces AV block, breaks re-entry).
 4. **Synchronized DC cardioversion (0.5–2 J/kg)** if adenosine fails.
- **Maintenance:** flecainide or sotalol. **Digoxin** if no delta wave; **propranolol** added if pre-excitation. 90% no further attacks after infancy → **stop treatment at age 1 if no WPW**. WPW relapse/risk → **radiofrequency/cryoablation** of accessory pathway.

Long QT syndrome

- Sudden LOC during **exercise/stress/emotion** (late childhood); may be misdiagnosed as **epilepsy**. Unrecognized → **sudden death from VT**.
- **Autosomal dominant**, many genetic causes; **channelopathy** (Na/K/Ca channels). Associated: **erythromycin**, electrolyte disorders, head injury.
- Assess anyone with FHx sudden unexplained death or syncope on exertion.

Rheumatic Fever

- Multisystem **autoimmune response to group A streptococcus**; children **5–15 yrs**; may progress to RHD.
- Higher in low/middle-income countries (19/100,000) — **household overcrowding**.

Infective Endocarditis

- All CHD (**except secundum ASD**), including neonates, at risk. **Highest risk = turbulent jet** (VSD, coarctation, aorta, PDA) or **prosthetic material**.

- **Presentation:** fever, anaemia/pallor, **clubbing (late)**, **splinter haemorrhages**, **splenomegaly**, **microscopic haematuria**, changing cardiac signs, neurological signs (cerebral infarction), retinal infarcts, arthralgia, necrotic skin lesions.
- **Diagnosis:** **multiple blood cultures BEFORE antibiotics**; **echo (vegetations)** confirms but can't exclude; raised acute phase reactants (monitor).
- **Most common organism:** α -haemolytic streptococcus (**Strep viridans**).
- **Treatment:** **high-dose penicillin + aminoglycoside, 6 weeks IV**; surgical removal if infected prosthetic material.

High-yield one-liners

- **Critical AS / coarctation in neonate = duct-dependent** → prostaglandin keeps PDA open.
- **HF:** $CO = HR \times SV$ (preload/afterload/contractility); infant = feeding difficulty + sweating + poor weight gain.
- **Kawasaki:** fever $\geq 5d$ + 4 of 5 (eyes, lips/strawberry tongue, hands/feet peeling, rash, cervical node); IVIG + aspirin; coronary aneurysms.
- **SVT = commonest childhood arrhythmia, 250–300 bpm**; adenosine = treatment of choice; **WPW = delta wave**.
- **Long QT = syncope on exertion, mistaken for epilepsy, AD channelopathy, sudden death.**
- **IE:** highest risk turbulent jet (**NOT secundum ASD**); **Strep viridans**; blood cultures before antibiotics; penicillin + aminoglycoside 6 wks.